

1 Power Analysis for Run-In Clinical Trials under AR(1) and
2 General Correlation Structures

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5 Abstract

6 **Background.** The companion Paper 1 of this compendium derived closed-form power
7 formulas for run-in and common close observations under compound symmetry. Com-
8 pound symmetry assumes all pairs of observations are equally correlated regardless of
9 temporal separation, an assumption that is empirically rejected in most longitudinal-trial
10 data. $\text{AR}(1)$, $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k-k'|}$, is the natural alternative.

11 **Methods.** We derive the variance of the treatment-effect estimator under $\text{AR}(1)$ via
12 two routes: (i) the Woodbury identity applied to the $\text{AR}(1)$ marginal covariance, and (ii) a
13 moment-based approach using the cross-covariance between run-in and treatment-period
14 means. The two routes target different estimands: the GLS variance of the slope-difference
15 effect and the variance of the simple phase-mean difference. The GLS derivation accom-
16 modates arbitrary J_0 run-in observations and arbitrary J_2 common close observations.

17 **Results.** A taxonomy of five correlation cases is presented, and change-score cor-
18 relations are derived in closed form under compound symmetry and $\text{AR}(1)$. Numerical
19 comparisons using parameters from Alzheimer’s disease neuroimaging trials show that,
20 when the two structures are matched at a common nominal correlation parameter ρ , the
21 $\text{AR}(1)$ sample-size estimate is smaller than the compound-symmetric estimate (by 4-36%
22 at $\rho = 0.5$), with the deviation growing as ρ increases. A narrow anticonservative regime
23 exists at large J_0 and small ρ , where compound symmetry understates the variance by at
24 most about 6%.

25 **Conclusions.** At a matched nominal ρ , the compound-symmetry assumption un-
26 derlying Paper 1 is generally conservative: it overstates the required sample size, and
27 increasingly so as the autocorrelation grows. The exception is a narrow regime with many
28 run-in observations and low autocorrelation, where it is anticonservative by at most about
29 6%. The $\text{AR}(1)$ formulas derived here quantify both effects and should be used directly
30 when serial correlation is expected; this conclusion is specific to the common-nominal- ρ
31 matching convention stated in the text.

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55 1 Introduction

56 [1] The efficiency of run-in designs is governed by the correlation struc-
57 ture of the repeated measurements.

- 58 • Under compound symmetry the correlation between change scores is con-
59 stant.
- 60 • Run-in observations then inform the random slope through simple averaging.
- 61 • Under AR(1) the correlation decays with separation, so run-in information
62 depends on distance from the treatment period.

63 *The efficiency of run-in designs depends on the correlation structure of the repeated mea-*
64 *surements. Under compound symmetry, the correlation between any two change scores is*
65 *constant, and the run-in observations contribute information about the random slope b_i*
66 *through a simple averaging mechanism. Under AR(1), the correlation decays with tem-*
67 *poral separation, and the information content of run-in observations depends on how far*
68 *they are from the treatment period.*

69 [2] This paper extends prior compound-symmetric run-in frameworks
70 to AR(1) residual correlation.

- Paper 1 assumed the rank-one-plus-identity change-score covariance from independent errors.
- Wang (2019) extended run-in to two-period mixed models but kept compound symmetry.
- We instead adopt AR(1) residual correlation with geometric decay in lag.

Paper 1¹ assumed $R = \sigma^2(I_J + \mathbf{1}_J \mathbf{1}_J')$, which arises from the change-score parameterization under independent errors.² extended the run-in framework to two-period linear mixed models but retained the compound symmetry assumption. We extend here to AR(1) residual correlation, where $\text{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|}$ for $0 < \rho < 1$.

[3] Existing literature treats autocorrelation and change-score variation but not the run-in setting.

- Winkens (2007) studied optimal designs for autocorrelated data outside the run-in context.
- Diggle and Verbeke give general treatments of longitudinal correlation structures.
- Munoz et al. (1992) embedded compound symmetry and AR(1) in a single damped-exponential family, the natural generalization of the two structures contrasted here.
- Chambless (1993) addressed intra-individual variation in change scores, a concern amplified under AR(1).

³ studied optimal designs for autocorrelated longitudinal data but did not consider the run-in setting.⁴ and⁵ provide general treatments of correlation structures in longitudinal models.⁶ proposed a damped-exponential family of correlation structures with $\text{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|^\theta}$, which contains compound symmetry ($\theta = 0$) and AR(1) ($\theta = 1$) as the two boundary members; the comparison developed here is thus a comparison of the two ends of that family, and the intermediate members offer a route to relaxing the binary choice.⁷ addressed the effect of intra-individual variation on change-score analyses, a concern that is amplified under AR(1) correlation.

[4] The closed-form AR(1) derivation covers a design case the reference longpower toolkit does not.

- Its slope-based closed-form routines assume independent or compound-symmetric residuals.
- Its `power.mmr.ar1()` routine implements the analytic AR(1) formula of Lu, Luo, and Chen (2008), but without run-in or common close observations under a random slope.
- Our formula supplies that case and collapses to the compound-symmetric structure of Paper 1 as $\rho \rightarrow 0$.

The longpower R package⁸ is the CRAN reference implementation for longitudinal-trial power calculations. Its `power.mmr.ar1()` routine implements the analytic AR(1) sample-size formula of⁹ for mixed models for repeated measures, and its slope-based closed-form routines (`edland.linear.power()`, `diggle.linear.power()`) assume random intercepts and slopes with independent or compound-symmetric residuals. None of these routines accommodates run-in or common close observations under a random-slope model. The

closed-form $AR(1)$ derivation developed here supplies that case. As $\rho \rightarrow 0$ (no serial correlation), the $AR(1)$ change-score covariance collapses to the compound-symmetric structure of Paper 1, so the present formulas contain the Paper 1 results as a special case.

2 Correlation structures for longitudinal trials

2.1 Taxonomy of cases

We consider five correlation structures for the raw outcomes Y_{ijk} :

1. **Compound symmetry:** $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \rho$ for all $k \neq k'$. This is equivalent to the random intercept model and produces the $R = \sigma^2(I + \mathbf{1}\mathbf{1}')$ structure in change scores.
2. **AR(1):** $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k'-k|}$. Correlation decays geometrically with lag.
3. **Compound symmetry with variable run-in:** Case 1 with $J_{0,i}$ varying across participants.
4. **AR(1) with variable run-in and common close:** Case 2 with $J_{0,i}$ and $J_{2,i}$ varying.
5. **General (unstructured):** $\text{Corr}(Y_{ijk}, Y_{ijk'}) = \sigma_{|k-k'|}/\sigma^2$.

Throughout, the design quantities J_0 , J_1 , and J_2 correspond to the arguments `r`, `k`, and `f` of the implementing `runinpower` functions.

2.2 Change-score correlations under each structure

Under the consecutive-difference parameterization ($C_j = Y_j - Y_{j-1}$, successive visits separated by t), the correlation between change scores depends on the raw-outcome correlation structure. Here $a = \sigma^2$ and $b = \sigma_b^2$. (The GLS machinery of the next section uses change scores from baseline; the two parameterizations are related by a nonsingular linear map, so the GLS variance is unaffected.)

For the run-in design with $J_0 = 2$, $J_1 = 1$, with visits at times $-2t, -t, 0, t$ and consecutive differences C_1, C_2, C_3 :

Compound symmetry (independent raw errors; the shared intercept component cancels in differencing):

$$\text{Corr}(C_1, C_3) = \frac{bt^2}{2a + bt^2} \quad (1)$$

$$\text{Corr}(C_1, C_2) = \frac{-a + bt^2}{2a + bt^2} \quad (2)$$

AR(1): for consecutive differences at lag $m = |j - l| \geq 1$,

$$\text{Var}(C_j) = bt^2 + 2a(1 - \rho) \quad (3)$$

$$\text{Cov}(C_j, C_l) = bt^2 - a\rho^{m-1}(1 - \rho)^2 \quad (4)$$

140 so that

$$\text{Corr}(C_1, C_3) = \frac{bt^2 - a\rho(1 - \rho)^2}{bt^2 + 2a(1 - \rho)} \quad (5)$$

$$\text{Corr}(C_1, C_2) = \frac{bt^2 - a(1 - \rho)^2}{bt^2 + 2a(1 - \rho)} \quad (6)$$

141 These expressions hold for arbitrary (J_0, J_1, J_2) because they depend only on the lag m ; at
142 $\rho = 0$ they reduce to the compound-symmetry formulas above.

143 3 Variance under AR(1): GLS approach

144 3.1 Marginal covariance

145 Under AR(1) with residual variance σ^2 and autocorrelation ρ , the raw-outcome covariance is
146 $\text{Cov}(e_{ij}, e_{ij'}) = \sigma^2 \rho^{|j-j'|}$. The change-score residual covariance becomes:

$$R_{jl} = \text{Cov}(e_{ij} - e_{i0}, e_{il} - e_{i0}) = \sigma^2(\rho^{|j-l|} - \rho^{|j|} - \rho^{|l|} + 1) \quad (7)$$

147 for $j, l \neq 0$.

148 [5] **The AR(1) change-score covariance loses rank-one structure but the**
149 **Woodbury reduction survives.**

- 150 • The matrix R is no longer rank-one-plus-identity, so Sherman-Morrison does
151 not apply.
- 152 • The Woodbury identity still reduces the inverse covariance to a scalar inver-
153 sion.
- 154 • This holds because the random-effects covariance G remains scalar.

155 *This R no longer has the rank-one-plus-identity structure, so the Sherman-Morrison for-*
156 *mula does not apply. However, the Woodbury identity still reduces Σ^{-1} to a scalar inversion*
157 *because G remains scalar.*

158 3.2 Woodbury computation

159 The computation proceeds identically to Paper 1:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \quad (8)$$

160 [6] **The Woodbury scalar term persists, but the residual inverse must**
161 **now be computed differently.**

- 162 • The correction term H remains a scalar as in Paper 1.
- 163 • The residual inverse R^{-1} must be obtained numerically.
- 164 • Equivalently it follows from the AR(1) tridiagonal inverse rather than
165 Sherman-Morrison.

166 *with $H = (G^{-1} + Z'R^{-1}Z)^{-1}$ still scalar. The difference is that R^{-1} must be computed*
167 *numerically (or via the AR(1) tridiagonal inverse) rather than by Sherman-Morrison.*

Table 1: Per-pair $\text{Var}(\hat{\gamma})$ under AR(1) for varying ρ ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	$\rho = 0$	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$
0	3.0000	2.9100	2.7500	2.5100	2.1900
2	2.1569	2.0132	1.6662	1.1157	0.3963
4	1.3268	1.3902	1.2218	0.8508	0.3036

4 Variance under AR(1): moment-based approach

4.1 Cross-covariance between phase means

[7] A moment-based alternative derives the pre-post variance directly from the phase means.

- It computes the variance of the test statistic from the moments of the phase means.
- The run-in mean averages the run-in observations.
- The treatment-period mean averages the treatment-period observations.
- The construction follows the summary-measures approach of Frison and Pocock (1992).

An alternative to the GLS approach computes the variance of the pre-post test statistic directly from the moments of the phase means, following the summary-measures approach of¹⁰, specialized here to AR(1) correlation. Let $\bar{X}_0 = \frac{1}{J_0} \sum_{i=1}^{J_0} Y_{i,run-in}$ and $\bar{X}_K = \frac{1}{J_1} \sum_{j=1}^{J_1} Y_{j,trt}$ denote the run-in and treatment-period means.

Under AR(1) with parameter $c = \rho$, the cross-covariance is:

$$\text{Cov}(\bar{X}_0, \bar{X}_K) = \frac{\sigma^2}{J_0 J_1} \rho c^{K-1} \frac{(1 - c^{J_0})(1 - c^{J_1})}{(1 - c)^2} \quad (9)$$

where $c = \rho$ (the symbol c here denotes the autocorrelation and is unrelated to the reparameterization $c = \sigma^2 - \sigma_b^2 t^2$ used in Paper 1) and K is the time-index lag from the last run-in visit to the first treatment visit on the visit grid $d = (-J_0, \dots, -1, 1, \dots, J_1)$. Because baseline (time 0) is not itself a phase observation, adjacent phases have $K = 2$. The division by $J_0 J_1$ reflects that $\bar{X}_0$ and $\bar{X}_K$ are means of J_0 and J_1 observations respectively; this matches the implementation in `ar1_cov_phase_means()` below.

4.2 Variance of the change score

The variance of the pre-post difference is:

$$\text{Var}(X_K - X_0) = \text{Var}(X_K) + \text{Var}(X_0) - 2 \text{Cov}(X_K, X_0) \quad (10)$$

The variance of each phase mean involves the sum $S_n = \sum_{i=1}^{n-1} (n-i)c^{i-1}$, which has the closed form:

$$S_n = \frac{n(1-c) - (1-c^n)}{(1-c)^2} \quad (11)$$

Moment-based vs GLS variance (r=2, k=2, rho=0.5):

```

194 ## Moment-based phase-mean difference: 1.2188
195 ## GLS slope difference (sigma_b2=0): 0.6364
196 ## GLS slope difference (sigma_b2=1): 1.6662
197 ## (Different estimands; see next section.)

```

198 5 Relationship between the two approaches

199 [8] **The GLS and moment-based approaches target distinct variance**
200 **quantities.**

- 201 • The GLS variance is the minimum-variance bound under the assumed model.
- 202 • The moment-based variance is that of the simple averaged change-score esti-
- 203 mator.
- 204 • The two therefore answer different efficiency questions.

205 *The GLS approach (Section 3) and the moment-based approach (Section 4) compute dif-*
206 *ferent quantities. The GLS variance is the minimum-variance bound under the assumed*
207 *model, while the moment-based variance is the variance of the simple averaged change-score*
208 *estimator $\bar{d}_1 - \bar{d}_0$.*

209 [9] **The two approaches do not coincide even without a random slope.**

- 210 • At $\sigma_b^2 = 0$ the GLS estimator still weights observations through the AR(1)
- 211 residual covariance and targets the slope difference.
- 212 • The moment-based quantity is the variance of the unweighted phase-mean
- 213 difference, a different estimand.
- 214 • Numerically, at $J_0 = 2$, $J_1 = 2$, $\rho = 0.5$, $\sigma^2 = 1$, $\sigma_b^2 = 0$: GLS 0.636 versus
- 215 moment-based 1.219.

216 *The two approaches do not coincide even when $\sigma_b^2 = 0$. At that boundary the GLS estimator*
217 *still weights the change scores through the AR(1) residual covariance and estimates the*
218 *slope-difference parameter γ , whereas the moment-based quantity is the variance of the*
219 *unweighted phase-mean difference $\bar{X}_K - \bar{X}_0$. With $J_0 = 2$, $J_1 = 2$, $\rho = 0.5$, $\sigma^2 = 1$, and*
220 *$\sigma_b^2 = 0$, the GLS variance is 0.636 while the moment-based variance is 1.219. When $\sigma_b^2 > 0$*
221 *the GLS estimator additionally exploits the random-slope structure of $\Sigma = R + ZGZ'$.*

222 [10] **The moment-based computation serves the simple pre-post estima-**
223 **tor, not as a bound on the GLS variance.**

- 224 • It requires no specification of the random-slope variance.
- 225 • It is the exact variance of the unweighted pre-post difference an analyst might
- 226 report without fitting a mixed model.
- 227 • Because the estimands differ, its ratio to the GLS variance is descriptive, not
- 228 an efficiency bound.

229 *The practical value of the moment-based computation is that it requires no specification*
230 *of σ_b^2 and gives the exact variance of the simple pre-post difference that an analyst might*
231 *report without fitting a mixed model. It is not a bound on the GLS variance: at the*
232 *parameters above the moment-based value (1.219) lies between the GLS variance at $\sigma_b^2 = 0$*

233 (0.636) and at $\sigma_b^2 = 1$ (1.666). Because the two quantities are variances of different
 234 estimands, their ratio describes the cost of abandoning the mixed model rather than an
 235 efficiency ordering within it. The relative-efficiency question is taken up in Paper 4.

236 6 Robustness to correlation misspecification

237 The practical question is whether a trial designed under compound symmetry loses substantial
 238 efficiency when the true correlation is AR(1), and vice versa. Throughout this section the two
 239 structures are matched at a common nominal correlation parameter ρ ; the comparison, and
 240 its direction, depend on this matching convention.

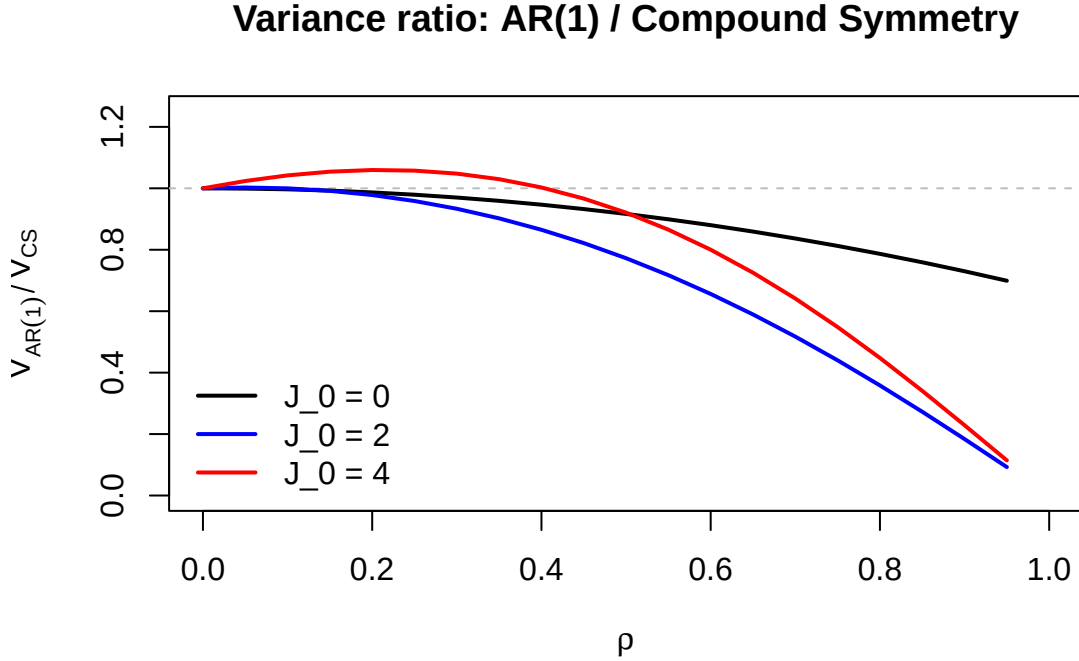


Figure 1: Ratio of AR(1) to compound-symmetry variance at a matched nominal ρ for three run-in configurations ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$). The ratio remains near or below one except at $J_0 = 4$ and small ρ , where it peaks near 1.06.

241 7 Stratified variance for unequal visit patterns

242 When participants have different numbers of run-in ($J_{0,i}$) or common close ($J_{2,i}$) observations,
 243 the total variance is a weighted sum across strata:

$$\frac{1}{n^2} \sum_{s=1}^S n_s \text{Var}(X_K - X_0) \big|_{J_0=J_{0,s}, J_2=J_{2,s}} \quad (12)$$

244 8 Numerical examples

245 8.1 Comparison across correlation structures

246 At the matched nominal $\rho = 0.5$, the AR(1) sample sizes are smaller than the compound-
 247 symmetric ones at every J_0 , by 3.6% ($J_0 = 0$) to 36.4% ($J_0 = 1$). Positive serial correla-

Table 2: Sample size per group (MIRIAD parameters, $\Delta = 0.25$, 90% power) under compound symmetry vs AR(1).

J_0	J_1	J_2	N (CS)	N (AR(1), $\rho = 0.5$)
0	2	0	385	371
1	2	0	247	157
2	2	0	145	107
3	2	0	100	85
4	2	0	78	72

tion reduces the residual variance of each change score ($R_{jj} = 2\sigma^2(1 - \rho^{|j|}) < 2\sigma^2$), so the compound-symmetric calculation, which corresponds to independent raw errors, is conservative here.

8.2 Sensitivity to rho

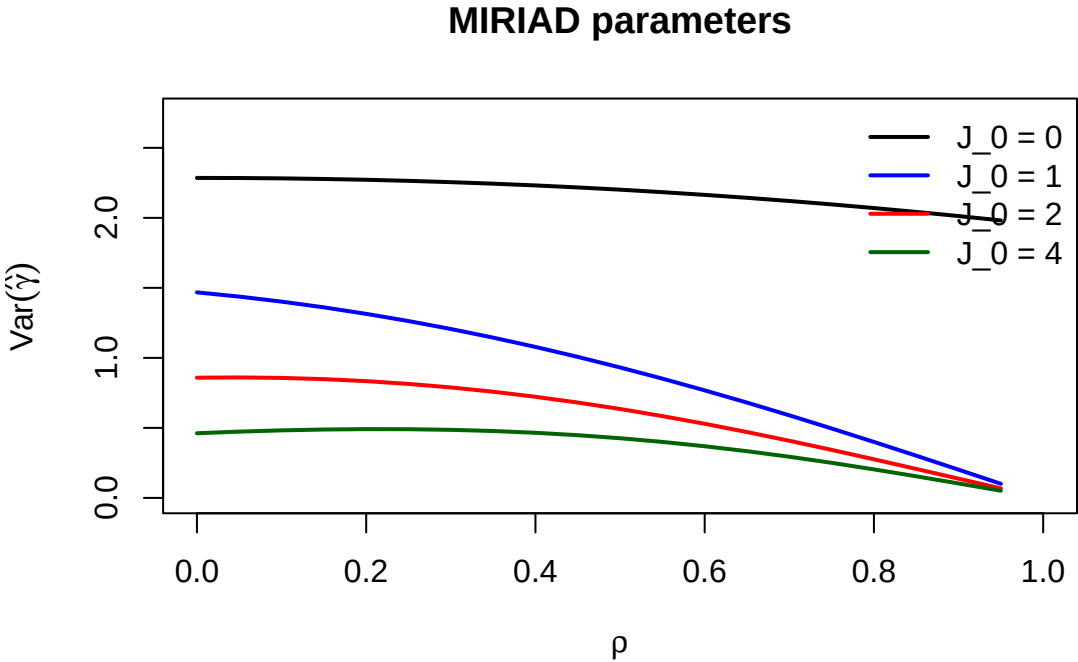


Figure 2: Per-pair variance as a function of ρ under AR(1) for MIRIAD parameters ($\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, $J_1 = 2$, $J_2 = 0$).

9 Discussion

[11] The autocorrelation structure matters for run-in designs only when the autocorrelation is weak relative to the random-slope signal.

- When the slope-to-noise ratio is large, the random-slope term dominates the change-score covariance and the residual correlation structure is second order.
- The AR(1) and compound-symmetry variances then differ negligibly, and the simpler Paper 1 formula suffices.

- The two structures diverge most when the slope-to-noise ratio is small and run-in observations are distant from randomization.

The central practical question is when the compound-symmetry formulas of Paper 1 can be used in place of the AR(1) formulas derived here. The numerical comparisons indicate that the answer is governed by the same slope-to-noise ratio $\sigma_b^2 t^2 / \sigma^2$ that governs the value of run-in observations themselves. When this ratio is large, the random slope term ZGZ' dominates the change-score covariance $\Sigma = R + ZGZ'$, the residual term R contributes little, and the choice between an AR(1) and a compound-symmetric R is immaterial. When the ratio is small, the residual term dominates and the correlation structure becomes the leading determinant of efficiency.

[12] **At a matched nominal ρ , compound symmetry is generally conservative, with a narrow anticonservative corner.**

- Positive serial correlation reduces the residual variance of every change score, so the AR(1) variance falls below the compound-symmetric value for most designs.
- The discrepancy grows with ρ ; near $\rho = 0.95$ the AR(1) variance can fall below one tenth of the compound-symmetric value.
- The exception is large J_0 at small ρ , where distant run-in observations lose information and compound symmetry understates the variance by at most about 6%.

The direction of the error is important for trial design. When the two structures are matched at a common nominal ρ , positive serial correlation reduces the residual variance of every change score ($R_{jj} = 2\sigma^2(1 - \rho^{|j|})$), so the AR(1) variance lies below the compound-symmetric value for most configurations and the compound-symmetric sample size is conservative. In the MIRIAD example at $\rho = 0.5$ the AR(1) sample sizes are 4 to 36 percent smaller than the compound-symmetric ones, and the gap widens as ρ grows. The exception is a narrow regime with many run-in observations and low autocorrelation ($J_0 = 4$, $\rho \approx 0.2$), where correlation decay makes distant run-in observations less informative than compound symmetry credits them to be; there the compound-symmetric formula understates the variance, by at most about 6 percent in the configurations examined. Both statements are specific to the common-nominal- ρ matching convention; matching the structures on, for example, a common lag-one change-score correlation would alter the comparison.

[13] **The numerical comparisons give a practical decision rule for when the simpler formula is safe.**

- At a matched nominal ρ , the compound-symmetry formula of Paper 1 is a conservative and convenient default for most designs.
- With many run-in observations and low expected autocorrelation, the AR(1) formula should be used directly; the compound-symmetric formula can understate the variance by up to about 6% there.
- When ρ is uncertain, evaluating the AR(1) formula over a plausible range of ρ and taking the largest variance is a safe default.

For applied use we recommend the following. At a matched nominal ρ , the compound-symmetry formula of Paper 1 is conservative for most designs and can serve as a convenient

303 *default; the AR(1) formula then refines the sample size downward. When the design places*
304 *many run-in observations before randomization and the expected autocorrelation is low,*
305 *the AR(1) formula should be used directly, because the compound-symmetric formula can*
306 *understate the variance by up to about 6 percent in that regime. When the correlation*
307 *structure cannot be pinned down from prior data, evaluating the AR(1) formula over a*
308 *plausible range of ρ and taking the largest resulting variance yields a safe sample size. The*
309 *moment-based computation of Section 4 addresses the simple pre-post estimator and is not*
310 *a bound on the GLS variance.*

311 10 Conflict of interest

312 The author declares no conflicts of interest.

313 11 Funding

314 This work received no specific funding.

315 12 Data availability statement

316 [14] **The implementing package, tests, and bibliography are openly avail-**
317 **able in the project repository.**

- 318 • The package implements the AR(1) variance formulas.
- 319 • Validation tests and the shared four-paper bibliography are included.
- 320 • Specific paths appear in the Reproducibility section below.

321 *The `runinpower` R package implementing the AR(1) variance formulas (`covariance-ar1.R`),*
322 *validation tests, and the shared bibliography across the four-paper compendium are openly*
323 *available in the project repository. Specific paths are listed in the Reproducibility section*
324 *below.*

325 13 Reproducibility

326 [15] **The manuscript was rendered with R 4.4.x using a small set of**
327 **principal packages.**

- 328 • The in-package source supplies the AR(1) machinery.
- 329 • Testing uses the `tinytest` harness.
- 330 • The manuscript build uses `rmarkdown` and `bookdown`.

331 *This manuscript was rendered with R 4.4.x. Principal packages used are the in-package*
332 *`runinpower` source (the AR(1) machinery is in `covariance-ar1.R`), `tinytest` (testing*
333 *harness), and `rmarkdown` plus `bookdown` for the manuscript build.*

334 [16] **All reported quantities are deterministic.**

- 335 • Given fixed inputs the variance formulas exercise no random-number genera-
- 336 tor.
- 337 • No Monte Carlo computation appears in this manuscript.

338 • Re-running the chunks therefore reproduces every table and figure exactly.

339 *Random-number generator.* The $AR(1)$ variance formulas are deterministic given
 340 fixed input parameters; no random-number generation is used anywhere in this manuscript,
 341 and re-running the code chunks reproduces every table and figure exactly.

342 [17] **All source, companion manuscripts, and bibliography reside at doc-**
 343 **umented repository paths.**

344 • The R package source is in the package source directory.

345 • The companion manuscripts for Papers 1, 3, and 4 have their own analysis
 346 subdirectories.

347 • This manuscript source and the shared bibliography reside in the correlation-
 348 paper directory.

349 *Data and code availability.* The R package source is at R/. The companion
 350 manuscripts (Papers 1, 3, 4) are at `analysis/report/{01,03,04}-*/`. The manuscript
 351 source and shared bibliography (`references.bib`, a symlink to `../..references.bib`)
 352 are at `analysis/report/02-correlation/`.

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